

Cyber Range Instructor Tools – Example Documentation

Overview

The Instructor Tools Panel provides a set of controls used by faculty and TAs during labs, exercises, and Capture-the-Flag (CTF) style scenarios. These tools allow the instructor to perform range management tasks such as resetting scores, restarting services, and refreshing telemetry.

In the current version, all actions are simulated for safety. In the final deployment, each action will trigger a backend API call to the appropriate Raspberry Pi (Router Pi, Server Pi, DMZ Pi, etc.).

1. Scoreboard Controls

1.1 Reset All Scores

Purpose: Clears all team points.

When to use: New round, debrief, testing.

1.2 Trigger Test Scoring Event

Purpose: Injects a synthetic scoring event.

When to use: Verify scoreboard, demonstrate event flow.

1.3 Freeze Scoring

Purpose: Temporarily pauses scoring.

When to use: Debrief, explanation break.

2. Network Controls

2.1 Restart Router Pi Network Stack

Purpose: Restarts routing/firewall services.

When to use: Nodes all down, network issues.

2.2 Restart DMZ Web Service

Purpose: Restarts vulnerable web service.

When to use: Reset DMZ target, service failure.

2.3 Reload Telemetry Snapshot

Purpose: Refresh node metrics.

When to use: Stale dashboard data.

3. Confirmation Modal

Explains action details and backend endpoint before execution.

4. Recent Action Log

Tracks last 5 instructor actions for reference.

5. Future Work

- Real APIs
- Live event feed
- Full range resets
- Link to lab PDFs